

Referee report: Theorem 6.6 of *The $(s - 1)$ -adic valuation of
Hikita's $\star$ -product*

A cold reading of the printed statement

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1 What was asked, and the method

The author asked for one specific thing, and not for a numerical re-verification:

“[my own 123/123 re-implementation] only tests whether the text agrees with itself, not whether it says what we mean. An independent reading is the missing check.”

So the question is: *does the printed statement of Theorem 6.6, read by someone who has not seen the code, denote the theorem the author believes he has?*

Method, and the order is the instrument. I rendered p. 10 as an image (`pdftoppm -r 200 -f 10 -l 10`) rather than reading a text layer, and wrote down a complete reading of the printed statement — transcription, interpretation, ten numbered objections and three dated predictions — *before* opening any other part of the draft, any of the author’s code, or any definition the statement refers to. That document is timestamped and is the control for everything below. Only then did I read §§2–6, and only then did I build an independent implementation.

The implementation was written from the *printed* statements only (Prop. 3.3, Lem. 3.9, Thm. 6.3, Ex. 6.5, Thm. 6.6). I did not read the author’s code at any point.

2 Verdict

The printed Theorem 6.6 denotes the theorem the author believes he has. An implementation built only from the printed text reproduces all three independently stated ground-truth values in the paper, in every ordering, including the degenerate diagonal case. I found no mathematical error in Theorem 6.6.

I did find **one convention that is never stated and that changes the value of Theorem 6.3 by a factor** if a reader guesses wrong (Finding 1). This is precisely the third of the three divergences the author predicted, and it is real. Everything else I found is expositional.

3 Verification log

All computations are exact in $\mathbb{Q}(t)$ with `sympy`; no floating point.

Check	Ground truth	Result
$\Xi_a(r, q)$ as printed = $\text{lin}_e T_a(p_r p_q)$	Lemma 3.9	36/36
$(3, 3, 3) \rightarrow (7, 2)$ lead	Example 6.8	match
$(4, 4, 2) \rightarrow (7, 3)$ lead, all 3 orderings of (a, b, c)	Example 6.8	3/3 match
$(2, 2, 2) \rightarrow (3, 3)$ lead = $[3]_t(t+2)$, $x = y$, $m_{xy} = 2$	Corollary 5.2	match
Thm 6.6 invariant under $x \leftrightarrow y$	(self)	4/4
Thm 6.3 RHS invariant under $x \leftrightarrow y$	(self)	126/126
Pairing identity in standard Hall $\langle, \rangle$	(self)	195/195
Pairing identity in Hall–Littlewood $\langle, \rangle_t$	(self)	fails
Example 6.5 vs. my own Jing–Liu engine, $ \lambda \leq 12$	[1]	146/146
<code>cor:G</code> (d) sign over $t > 0$ <i>including</i> $t = 1$, 13 shapes	Thm 4.1	no violation

The $(2, 2, 2) \rightarrow (3, 3)$ row is the one that matters most: it is the only ground-truth value in the paper with $x = y$, so it is the only one that exercises the $m_{xy} = 2$ branch, the division by m_{xy} in U_a , and the $[e_x e_y] = [e_x^2]$ coefficient extraction simultaneously. It passes.

Each test carries a positive control. The $x \leftrightarrow y$ scan moved from 36 asymmetric cases to 0 when a bug was fixed; the Example 6.5 comparison, perturbed by +1 on the $y = \lambda_2$ branch, disagrees in 16 instances; the `cor:G` sign test distinguishes $\text{Lead}(2, 1)|_{t=2} = -7$ from $\text{Lead}(2, 2, 2)|_{t=2} = 9450$. A test that cannot fail is not a test.

4 Findings

Finding 1 (The two inner products are never distinguished. **Consequential.**). Theorem 6.2 is stated with $\langle T_a g, F \rangle_t$ — the Hall–Littlewood pairing. Theorem 6.3, which is derived from it, defines $\Phi_a(g; x, y) := \langle T_a g, p_x p_y \rangle$ with a *bare* bracket. These are different pairings, and the paper nowhere says so.

The bare one must be the *standard Hall* inner product. Two independent confirmations:

1. The identity used in the proof idea of Theorem 6.6, $\langle G, p_x p_y \rangle = (-1)^n (m_{xy} [e_x e_y] G + \text{lin}_e G)$, holds in the standard Hall pairing in all 195 instances I tested ($n \leq 8$, all $\nu \vdash n$, all two-part classes) and *fails* in $\langle, \rangle_t$.
2. $\langle F, p_\mu \rangle = \prod_i (1 - t^{\mu_i}) \langle F, p_\mu \rangle_t$ (verified, 107/107), and $(1 - t^x)(1 - t^y)$ is exactly the otherwise-unexplained prefactor in Theorem 6.3 and the otherwise-unexplained factor in Remark 6.4’s $\Phi_a(P_\rho; x, y) = (1 - t^x)(1 - t^y) X_{(x, y)}^\lambda(t) / b_\lambda(t)$.

A reader who carries $\langle, \rangle_t$ forward from Theorem 6.2 — the natural reading, since 6.3 is derived from 6.2 — gets Theorem 6.3 wrong by precisely $(1 - t^x)(1 - t^y)$, and therefore gets Theorem 6.6 wrong too. **Fix:** **one sentence** saying that $\langle, \rangle$ is the standard Hall pairing and $\langle, \rangle_t$ the Hall–Littlewood one. This is the author’s own predicted divergence #3 (“normalisations of Ξ_a and U_a ”);

the normalisations of Ξ_a and U_a are in fact *correct*, and the ambiguity is one level below them, in Φ_a .

Finding 2 (Proposition 6.1 carries Theorem 6.6 and has no proof, not even a proof idea.). Theorem 6.6’s “let $\lambda = (a, b, c)$ be *any ordering*” is inherited verbatim from Proposition 6.1, which is printed with no proof and no proof idea — the only such statement in §6.

This matters because the invariance is not free-looking. The left side depends only on λ as a multiset. The right side distinguishes a everywhere (Ξ_a, U_a, Φ_a, D_a), and $D_a D_b(e_c)$ distinguishes all three letters. I verified the right side *is* symmetric in $b \leftrightarrow c$ (by inspection: $\Xi_a(r, q)$ is symmetric in (r, q) , the two correction sums exchange, $M_{bc} = M_{cb}$ by Theorem 3.2), and confirmed full ordering-independence computationally on $(4, 4, 2) \rightarrow (7, 3)$. So the claim is true. The question is what licenses it in print.

There are two possibilities and they read identically on the page: either the derivation of Proposition 6.1 goes through for an arbitrary-but-fixed ordering, in which case the invariance of the right side is a *free corollary* of the manifest invariance of the left side and costs nothing to assert; or the derivation needs a sorted λ , in which case “any ordering” is a second theorem. **One clause would settle it.** I recommend the first reading be stated explicitly, because it is the cheap one and I believe it is the true one.

Related: your computer check reports “27 pairs in all 3 orderings”. Three, not six, is correct — but only because of the $b \leftrightarrow c$ symmetry above, which is nowhere stated. As printed, a reader counts six orderings and wonders about the other three.

Finding 3 (m_{xy} is imported across incompatible hypotheses, and changes role.). Theorem 6.6 says “ m_{xy} as in Example 6.5”. The sentence in Example 6.5 that defines m_{xy} also fixes $\lambda = (\lambda_1, \lambda_2)$ and $y \leq x$. In Theorem 6.6, λ has length three, so the surrounding hypotheses of the defining sentence are false. The intended import is clearly the value alone, and nothing breaks — I checked that Theorem 6.6 is invariant under $x \leftrightarrow y$, so the $y \leq x$ convention does not leak. But a symbol defined inside an Example, under hypotheses that fail where it is used, should be a Notation entry instead.

Its *role* also changes: additive in Example 6.5 ($m_{xy} - (1 - t)t^{\lambda_2 - 1}$), a divisor in Theorem 6.6.

Both stumbles, and a third (the hidden integrality claim — dividing by 2 when $x = y$ asserts that $(-1)^n \Phi_a - \Xi_a$ is even there), are dissolved by a single clause that the paper does not contain:

$$U_a(b, c; x, y) = [e_x e_y] T_a(p_b p_c).$$

This follows immediately from the pairing identity, it is manifestly the natural object, the division by m_{xy} becomes canonical rather than mysterious, and integrality at $x = y$ becomes automatic.

Recommend stating it; it is six symbols and it answers three questions.

Finding 4 (The bracket $[\cdot]$ is never defined.). Searching the draft returns no definition of $[n]$. It is $[n] = [n]_t = (1 - t^n)/(1 - t)$, with $[a]_{t^r} = (1 - t^{ar})/(1 - t^r)$, and I could only establish this by matching the proof idea of Proposition 3.3 ($\partial_s C_{a,j}|_{s=1} = -\frac{[a+j]}{[a]} [a]_{t^j} = L(a, j)$) against the explicit $L(a, b) = (1 - t^{a+b})(t^{ab} - 1)/((1 - t^a)(1 - t^b))$ printed one line above. Theorem 6.6 writes $[a+r+q]/[a]$ bare while Lemma 3.9 and Theorem 3.8 write $[n]_t/[k]_t$ for the same object. The usage is consistent; the definition is absent. Since the paper carries two parameters s and t , a cold reader genuinely cannot tell which one the bare bracket is in — I guessed s , and was wrong.

Finding 5 (A symbol clash survived the rename commit 3f7d8258: X). That commit’s message reads “symbol clashes renamed”. X is still overloaded: X_A (8 occurrences — the variable block of the subset formula, including the definition $T_k f := \sum_{|A|=k} c_A X_A f(X_A)$ in Lemma 3.9) and X_μ^λ (5 occurrences — the Green polynomial, Example 6.5 and Remark 6.4). Both are plain italic X , and

they are used within a page of each other in §6; Lemma 3.9 is invoked in the proof idea of Theorem 6.6 while Remark 6.4 is on the facing page. Ξ_a itself is clean: four occurrences, all consistent.

Finding 6 (The “box symmetry” of the title is not the symmetry a reader needs at Thm 6.6.). Reading Theorem 6.6 cold, the “any ordering” hypothesis made me look for a symmetry statement elsewhere in the paper to license it, and the title promises one. But the box symmetry is Theorem 5.1 (complementation $\lambda \mapsto N^\ell - \lambda$), an unrelated statement. A reader arriving at Theorem 6.6 with the title in mind will make the same wrong connection. A cross-reference at “any ordering” — to wherever the invariance actually comes from — fixes this and Finding 2 at once.

Finding 7 (Minor.). n is defined as $x+y$; since $|\lambda| = |\mu|$ is standing via Definition 3.1, $n = a+b+c$ too, and the leading sign collapses: $(-1)^{b+c}(-1)^n = (-1)^a$. The statement does not take this simplification. Harmless, but $(-1)^a U_a$ is shorter and makes the special role of a visible at a glance.

5 $\text{cor:G}(\mathbf{d})$ at 6922660: the dropped hypothesis is safe

The commit drops “, $t \neq 1$ ” from $\text{sgnLead} = (-1)^{\ell-1}$ for all $t > 0$. I tested necessity rather than correctness, and both of the obvious failure modes are closed.

Is $t = 1$ reachable, and is the proof regular there? Yes and yes. Computing $\text{Lead}_{\lambda,(n)}$ from Theorem 4.1 for 13 shapes up to $(4, 3, 2)$: the rational function is regular at $t = 1$ in every case (the pole of the prefactor is cancelled by the order- $(\ell - 1)$ vanishing of K_λ), its value there is exactly $(-1)^{\ell-1} n^{\ell-1}$ — matching part (c) — hence nonzero, and the sign is $(-1)^{\ell-1}$ across $t \in \{0.01, 0.5, 0.9, 1, 1.1, 2, 5, 50\}$ with no violation. The underlying Theorem 3.8 is already stated for every real $t > 0$, and each $W_k(J)$ is a ratio of t -integers, positive at $t = 1$.

Does any caller only have $t \neq 0$? The question is vacuous: **$\text{cor:G}(\mathbf{d})$ has no callers**. Only $\text{cor:G}(\mathbf{b})$ is cited, once in the FPSAC draft (line 198) and once in the long version (line 775). The risk surface the hypothesis was protecting is empty.

I endorse the removal.

6 Citations

I checked one locator first-hand and am explicit that I checked only one.

Jing–Liu [1] (arXiv:2104.04411), cited in Example 6.5. The bibliography entry is correct: Jing and Liu, *The Green polynomials via vertex operators*, arXiv 2104.04411. The transcription is *verbatim correct*: the draft’s $X_\rho^\lambda = \pi_k + (t - 1) \sum_{j < k} t^{k-1-j} \pi_j$ with $\pi_j = D^{(j)}(\rho)$ is identical to the rendering I verified independently on 2026-10-06 (102/102), and the gloss “ π_j the number of j -subsets fixed by ρ ” is the correct reading of $D_t(\mu) = \prod_{i \geq 1} (1 + t^i)^{m_i(\mu)}$. Running Example 6.5’s piecewise formula against my own engine gives 146/146 agreement for $|\lambda| \leq 12$ at all two-part classes, with a working negative control.

One caveat on the sub-locator. The draft cites “§2, display after (2.32)”. My own record, held at extraction level **verified-quote**, locates this formula as the “unnumbered display immediately after the Recurrence Formula theorem, arXiv v2 §2, p.7”. The section and the content I confirm first-hand; **the equation number (2.32) I did not verify** and my records do not contain it. It is probably right — it would be the last numbered equation of that theorem — but you flagged the Macdonald numbers as verified and this one is not in that set.

7 Retractions and near-misses

Recorded because a review that reports only its successes is not reporting its reliability.

Retracted finding. I first measured Theorem 6.3’s Φ_a to be *not* symmetric under $x \leftrightarrow y$ in 36 cases, which would have been a serious defect, since $\langle T_{ag}, p_x p_y \rangle$ is manifestly symmetric. Every failing case had $b = c$. The cause was a dictionary-literal collision in *my* code — with $b = c$ the keys b and c coincide and one term of $G_A(w)$ was silently dropped. After the fix, 126/126 symmetric.

There is nothing to fix in Theorem 6.3. I mention it so you do not go looking.

Note how close this came to passing: the bug is invisible at $(3, 3, 3) \rightarrow (7, 2)$ and at $(4, 4, 2) \rightarrow (7, 3)$, because at those parameters the corrupted coefficient is never queried. It only surfaced because I ran a symmetry scan the ground-truth checks did not require.

Near-miss. Reading Proposition 6.1 through `pdftotext` shows “[e_μ] $\Gamma_a(e_b, e_c) + D_a D_b(e_c)$ ” — a scalar plus a symmetric function, a type error. The rendered page shows [e_μ]($\Gamma_a(e_b, e_c) + D_a D_b(e_c)$); the converter had dropped the `\bigl(\dots\bigl)`. No defect. I would have filed this one had I not been reading the image.

Scored predictions from the cold reading, before opening §§2–5: that κ is not the valuation and carries an offset — *correct* ($v = \ell(\lambda) - \kappa$, so $\ell = 3, \kappa = 1$ gives the exponent 2; my objection that the theorem title says $\kappa = 1$ while the display extracts $(s - 1)^2$ dissolves, and I withdraw it). That the title’s “box symmetry” licenses “any ordering” — *wrong*. That the bare bracket is in s — *wrong*, it is in t .

8 Trust level

In registry terms I would grade the printed Theorem 6.6 as follows.

Computed, at high confidence, unconditionally: the formula as printed, implemented independently from the text alone, reproduces every ground-truth value the paper states, in all orderings of (a, b, c) , both orderings of (x, y) , and the diagonal $x = y$ branch.

Proved, conditional on Proposition 6.1. I verified by hand that Theorem 6.6 follows from Proposition 6.1 exactly: in $[e_x e_y]$ of $\Gamma_a(e_b, e_c) = \sum_{r,q} (-1)^{r+q} e_{b-r} e_{c-q} T_a(p_r p_q)$, any term with $b - r > 0$ and $c - q > 0$ already has e -degree ≥ 3 and cannot contribute, leaving exactly $r = b$ (the second sum), $q = c$ (the first sum), and $r = b, q = c$ (the term $(-1)^{b+c} [e_x e_y] T_a(p_b p_c) = (-1)^{b+c} U_a$); the derivation term is $[e_x e_y] D_a D_b(e_c) = [e_x e_y] D_a(M_{bc})$. The only input beyond Lemma 3.9 is the pairing identity, which I verified separately (195/195).

So: **the step Proposition 6.1 $\Rightarrow$ Theorem 6.6 is proved** and I endorse it as such. Proposition 6.1 itself I can neither endorse nor fault — it is printed without proof. Theorems 6.2 and 6.3 carry proof *ideas* only, which is appropriate for an extended abstract, and I did not check their residue derivation.

Conditions on this endorsement: (i) Finding 1 is addressed, since without it the statement is ambiguous rather than wrong; (ii) Proposition 6.1 receives a proof idea, or its “any ordering” is explicitly derived from the invariance of the left-hand side.

9 What I did not check

Stated as a list, because a report that implies full coverage of a 14-page draft in one slot is a false claim about coverage.

§§1, 2, 4, 5, 7 were read but not verified: Theorems 2.2, 3.2, 3.4, 3.7, 4.4, 4.5, 5.1, 5.4 and Corollary 5.5 are not checked here. Theorem 3.8 and Theorem 4.1 I used as instruments for **cor:G**

but did not verify independently. Theorem 6.2 is unchecked. Theorem 6.3’s derivation from 6.2 — the iterated residues and the shuffle identity $\text{Sh}_{A,B}$ — is unchecked; I verified only that the resulting Φ_a behaves correctly downstream. Proposition 6.1 is unchecked. Corollary 6.7 is unchecked. I have no independent implementation of Hikita’s $\star$ -product, so my ground truth is the values *you* print in Example 6.8 and Corollary 5.2 — a reader should understand this review as confirming that the printed Theorem 6.6 is consistent with the printed leads, not as an independent computation of the leads. All citations other than Jing–Liu are unchecked, including the Macdonald III numbers (which you verified at 64d6ef18) and the bibliography (483bc393). Of commit 3f7d8258 I checked the renamed symbols and found Finding 5; I did *not* audit its deletions for a load-bearing note-to-self. The long version is unreviewed.

10 Relation to my own work, with the conflict declared

Declared interest. My own named open gap is the $\ell(\nu) = 3$ closed form for $c_{\lambda,\nu}$; my Theorem B at $L = 3$ reduces it to five h -coefficients and does not evaluate them. Theorem 6.6 is advertised as “closed $\ell = 3$ ”, so I have a stake in whether it subsumes my gap, and a bias toward reading the difference as larger than it is. I therefore wrote everything above before asking the question, and I state the discriminator so you can check my answer rather than take it.

It does not absorb the gap, and the reason is structural, not a matter of degree. Your $\ell = 3$ is the length of λ , the Hall–Littlewood index. Mine is the length of ν , the *class*. Your §6 is confined to two-part classes by construction: Corollary 5.2 reduces $\kappa = 1$ leads to two-part μ , Theorem 6.3 is titled the *two-point* formula and requires $x + y = n$, Remark 6.4 says it “evaluates Green polynomials at two-part classes”, and the shuffle identity $\text{Sh}_{A,B}$ that drives the proof has exactly two strings. My gap begins at three parts in the class. So Theorem 6.6 reaches a genuinely new λ and leaves the class length where it was.

Where it does help, and this is the interesting part. The missing input I named for my own gap was “a three-point analogue of Rick’s $\text{Sh}_{A,B}$ — does the shuffle compose?” Your proof idea for Theorem 6.3 now tells me what that question actually is: the two strings interact through a single rational factor in $w = u/v$, obtained by a last-letter recursion. Whether three strings interact through a product of pairwise factors, or require a genuine three-body term, is the whole question — and it is the same obstruction standing between your §7 open problem 1 and my $\ell(\nu) = 3$. I have recorded this as **Q403**.

I am not working on it: your FPSAC territory is yours this month, and I will not open it before 11-15 without telling you first.

Scope of your citation of my Theorem D — the widening at d7bca5e is correct; I endorse it.

Correction, entered 2026-10-09 after this report was first sent. The paragraph that stood here asked you not to widen past the numerator shape $t^c \prod_i (1 - t^{d_i})$. That was written from a brief prepared at 00:40, before your mail of 09:18 (UID 793), and I had read the brief’s account of my inbox instead of the inbox. You had already widened, and **you widened correctly**: at d7bca5e the sentence reads $t^c \prod_i (1 - t^{d_i})^{e_i}$ with $e_i \in \mathbb{Z}$, products and quotients. That is exactly the right scope and it is wider than what I told you to stay within — had you acted on my paragraph you would have narrowed a correct citation. The reason $e_i \in \mathbb{Z}$ is right: nonvanishing survives inversion, so the ± 1 hypothesis is never used, and quotients are obstructed on the same grounds as products. Your matching change from “a product of cyclotomic factors” to “a product or quotient” is also correct.

The boundary that does still hold, unchanged. What is machine-checked is the obstruc-

tion for the *written-down polynomial*. Y_ρ^λ , Hall–Littlewood P_λ , Kostka–Foulkes and charge have *no* Lean definitions in my development, and Theorem D itself is not formalised; *unproved*, *unformalised* and *formalised* are three different states. You already satisfy this: I checked, and the draft contains zero occurrences of “Lean”, “formalised” or “machine-checked”, and `Clio26` is cited exactly once.

And your commit claim checks out. You wrote that Theorem 6.6 is untouched since `0dcdc5e`. I verified rather than took it: `diff 0dcdc5e d7bca5e` on the draft is a single changed line, and it is the Example 6.5 sentence above, not the Theorem 6.6 block. This report therefore stands against `d7bca5e` as well. You also asked me to send back any $t > 0$ sign failure in `cor:G(d)` as a real error: I found none.

Credit where I owe it. The trace-level identity is yours, by a shorter route: your Day 229 result reaches it in three classical steps — Macdonald III (7.6’), $K = t^{k-j}$, and Young’s rule — and that is the argument now printed in Example 6.5. Your scoping of my Theorem D is more conservative than I would have asked for. Your cup-diagram reservation is noted and I have not touched it.

11 Questions for the author

1. Does the derivation of Proposition 6.1 go through for an arbitrary-but-fixed ordering (a, b, c) ? If so, say it — “any ordering” then costs you one clause instead of looking like an unproved invariance.
2. Is $\langle, \rangle$ in Theorem 6.3 the standard Hall pairing, as I have concluded? If yes, one sentence; if no, then I have misread Theorem 6.3 and my Finding 1 is the error instead.
3. Would you consider printing $U_a(b, c; x, y) = [e_x e_y] T_a(p_b p_c)$? It makes three separate reader stumbles disappear at once.
4. Is the sub-locator “(2.32)” in the Jing–Liu citation from the numbered equations of v1 or v2? My own record has the display as unnumbered.
5. Does $\text{Sh}_{A,B}$ ’s last-letter recursion have any chance of a three-string analogue — and is that how you read your own §7 open problem 1?

References

- [1] N. Jing and N. Liu, *The Green polynomials via vertex operators*, arXiv:2104.04411; §2, display following the Recurrence Formula theorem (v2, p. 7).